SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Constituent of Symbiosis International (Deemed University), Pune
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COMPUTER SCIENCE AND ENGINEERING
BATCH 2022-26

MACHINE LEARNING

Title: **Global COVID-19 Vaccination Analysis & Impact on Excess Mortality**

GitHub Link: https://github.com/DevanshuSawarkar/Global_COVID19_Vaccination_Analysis

Course Name: **Machine Learning**

Course Code: **T7529**

Semester: **VII**
Section: **C**

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Project Aim & Objectives

Aim: To analyze global vaccination trends and quantify the statistical relationship between vaccination strategies (**"Value" vs. "Pace"**) and **excess mortality** in 2021.

Key Objectives:

1. Clean, prepare, and merge global vaccination and excess death datasets.
2. Explore trends and patterns in vaccination rollouts (EDA).
3. Model and predict vaccination campaign success using a suite of machine learning models.
4. Discover natural country groupings using unsupervised clustering.
5. Forecast vaccination pace using deep learning.
6. Quantify the specific impact of vaccination on reducing excess deaths.

Datasets

Dataset 1: Global COVID-19 Vaccinations

Source: Our World in Data (vaccinations.csv)

Data: 196k+ daily records on doses, people vaccinated, boosters, and pace.

Dataset 2: WHO Excess Mortality

Source: World Health Organization (WHO_COVID_...xlsx)

Data: 2020-2021 excess death estimates, aggregated by country.

Key Challenge: These datasets must be cleaned, merged, and normalized to be useful.

The Foundation: Data Cleaning & Normalization

1. **Handling Missing Data:** Used **forward-fill (ffill)** on vaccination data. This is critical for cumulative data to avoid incorrectly resetting counts to 0.

2. **Deriving Population:** Calculated population for each country using the formula: Population = (Daily Vaccinations / Daily Vax per Million) * 1,000,000

3. **Normalization (The Key Step):** Created our target metric, **excess_deaths_per_million**, by dividing total 2021 excess deaths by the derived population.

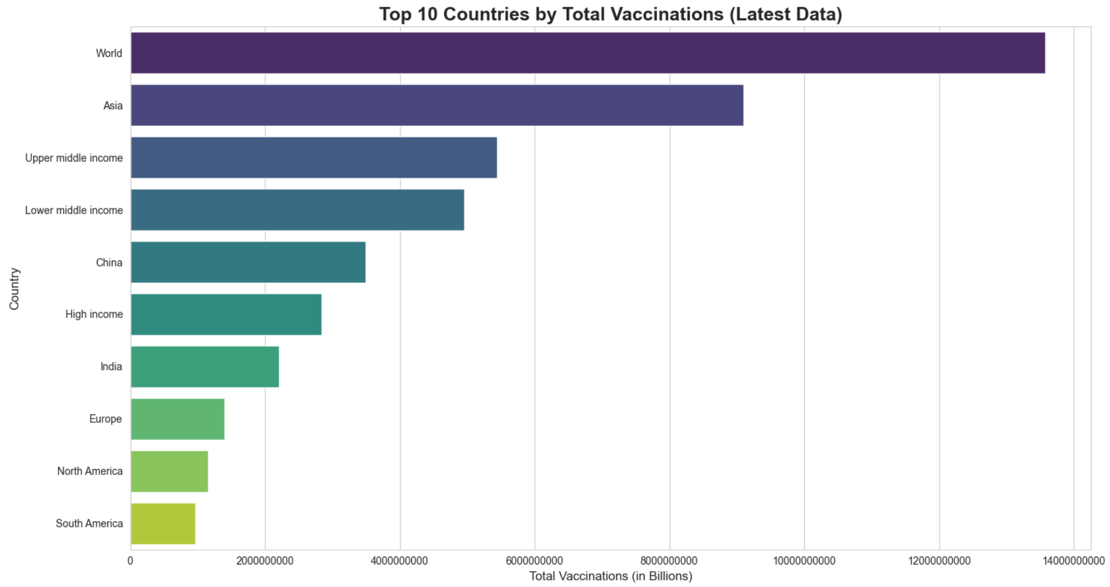
Result: A clean, merged dataset (vax_vs_deaths_2021_normalized.csv) that allows for a fair, "apples-to-apples" comparison between countries of all sizes.

Foundational Insights: EDA

Reasoning: To understand the basic trends and scale of the vaccination rollout.

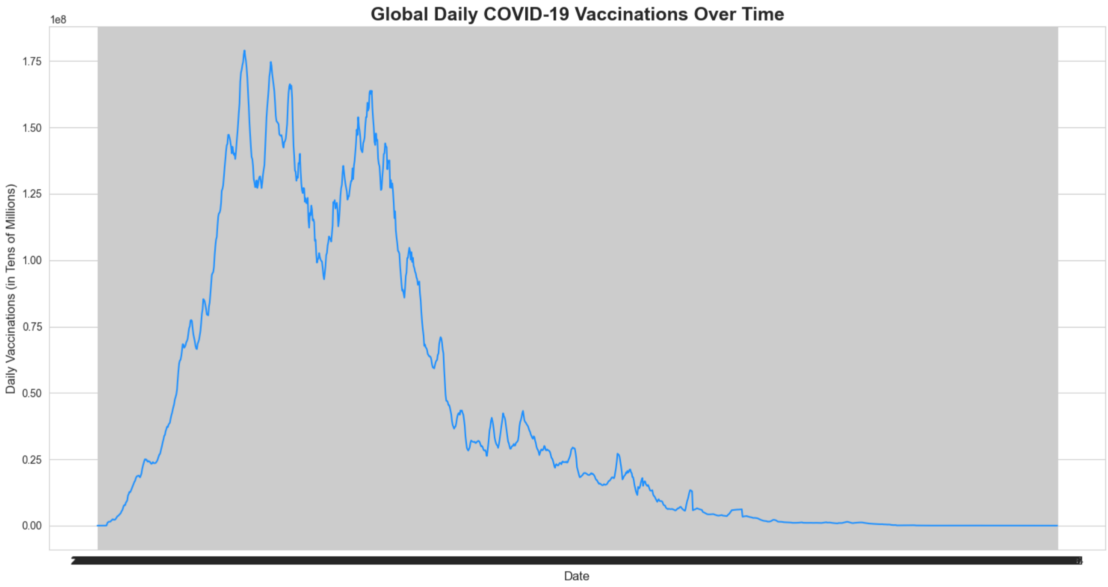
Graph 1:

Insight: The sheer scale of campaigns in China and India dwarfs all other nations.



Graph 2:

Insight: The global rollout was not a smooth curve but occurred in distinct **waves**, reflecting supply, variants, and booster pushes.

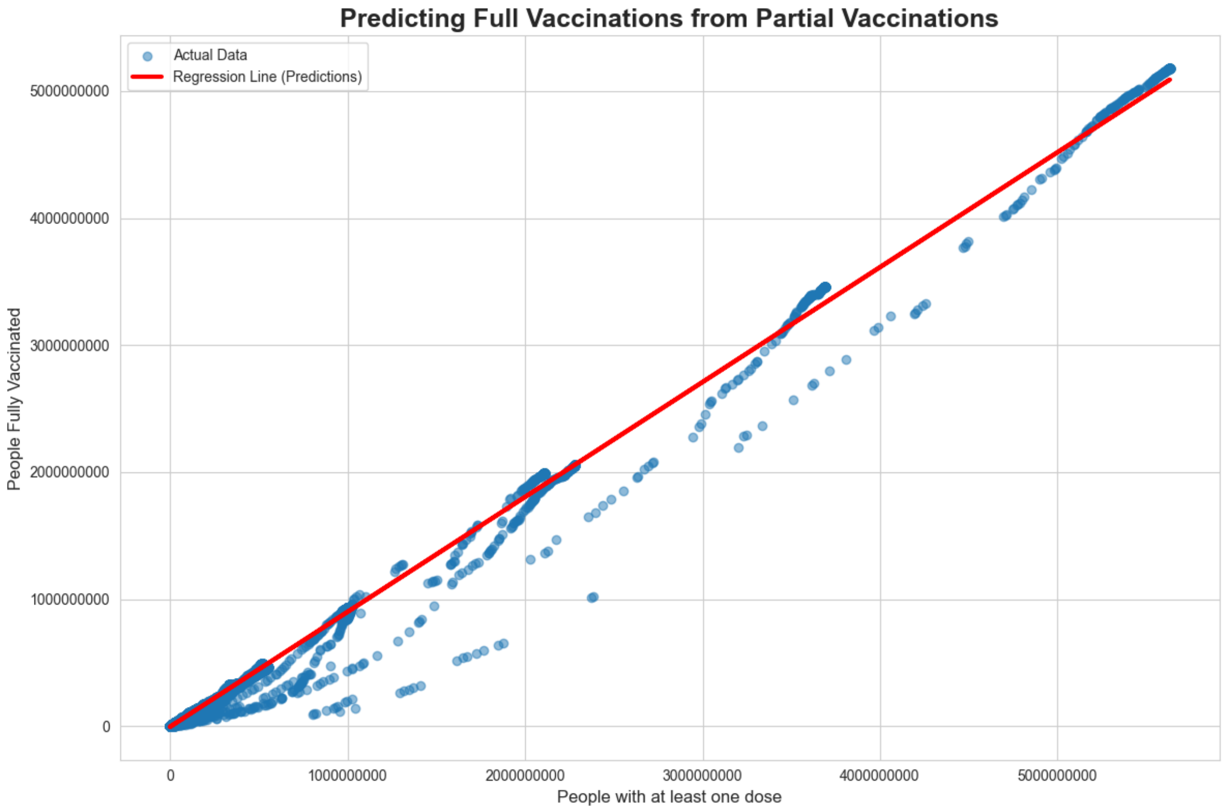


Simple Linear Regression

Aim: To predict the number of people_fully_vaccinated based on the number of people_vaccinated (at least one dose).

Reasoning: This helps us understand the "conversion rate" from a first dose to a full vaccination.

Graph:



Finding: A near-perfect R-squared of **0.98+**. This shows a highly reliable "conversion rate," which is extremely useful for logistical planning of second doses.

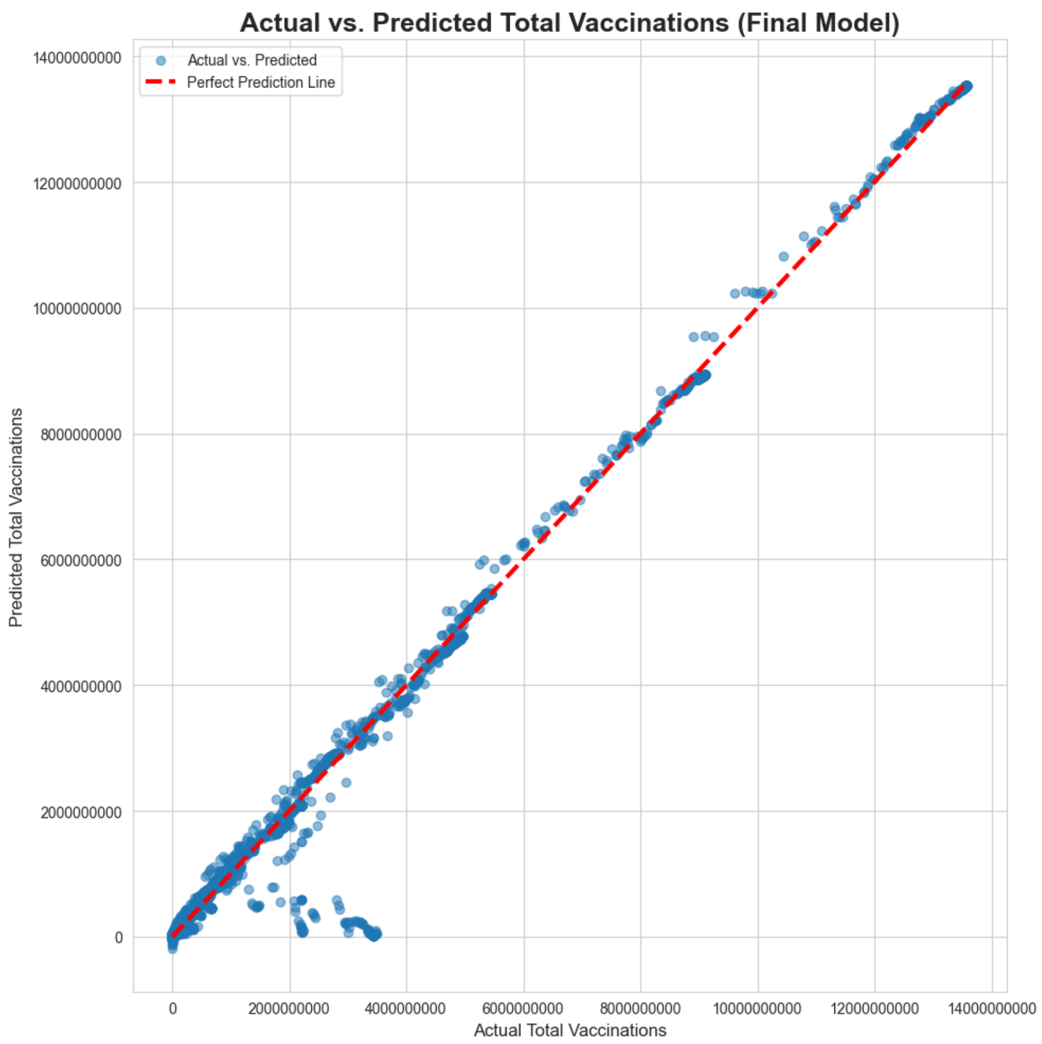
Multiple Linear Regression

Aim: To predict total_vaccinations using all other relevant metrics.

Reasoning: To test the integrity of the data and build a predictive model.

Method: Used **Backward Elimination** to remove redundant features (e.g., daily_vaccinations_per_million) and build a statistically sound model.

Graph:



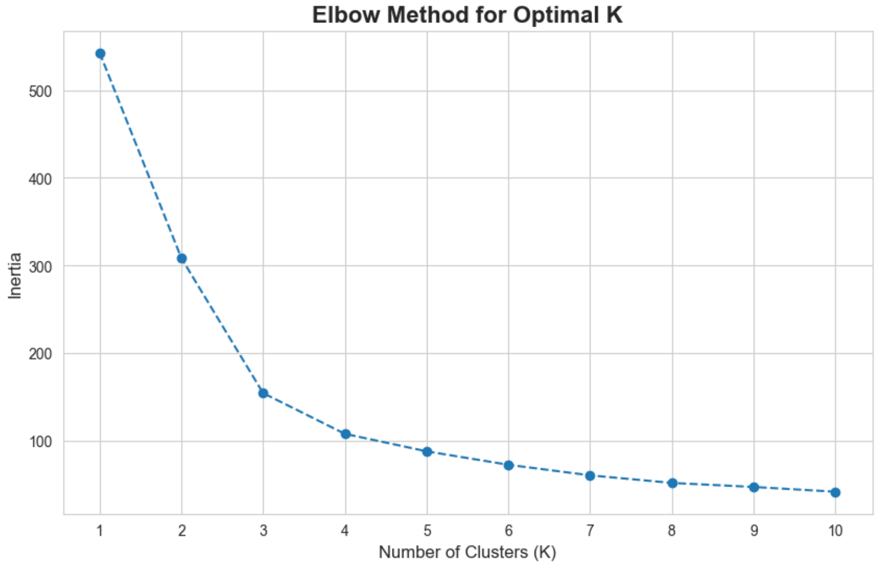
Finding: An R^2 of **0.999+** confirms extremely high data integrity. The model's predictions are nearly identical to the actual values.

K-Means Clustering (Unsupervised)

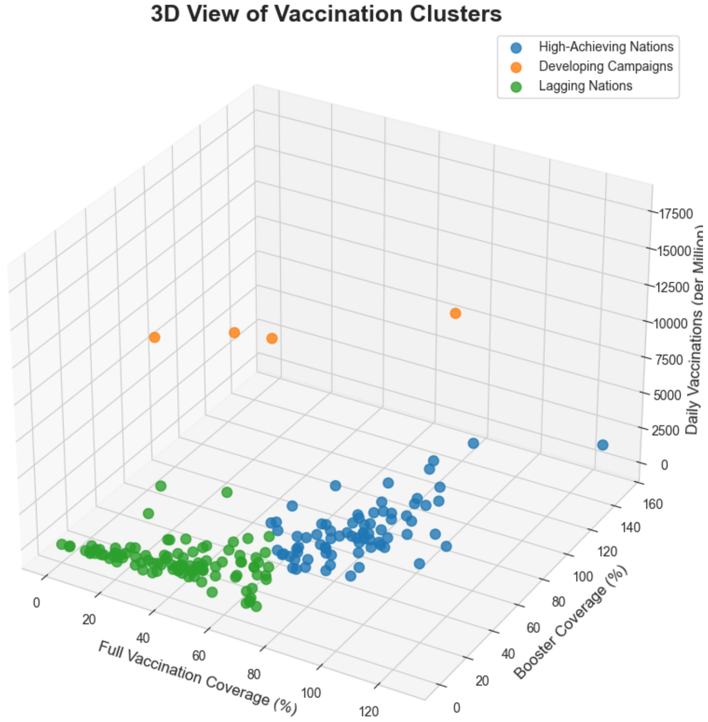
Aim: To find natural, hidden "archetypes" of countries based on their vaccination performance, without any predefined labels.

Method: Used the **Elbow Method** (Graph 1) to find the optimal number of clusters (K=3).

Graph 1:



Graph 2:



Finding: The model found 3 distinct groups:

1. **"High Achievers":** High coverage, mature campaigns.
2. **"Active Campaigns":** Moderate coverage, *very high* daily pace.
3. **"Lagging Nations":** Low coverage and low pace.

Predicting Vaccination Success

Aim: To classify countries as "High" or "Low" (>= 60% fully vaxxed) based on their *strategy*.

Features (The "Strategy"):

- total_boosters_per_hundred
- daily_vaccinations_per_million
- campaign_duration_days

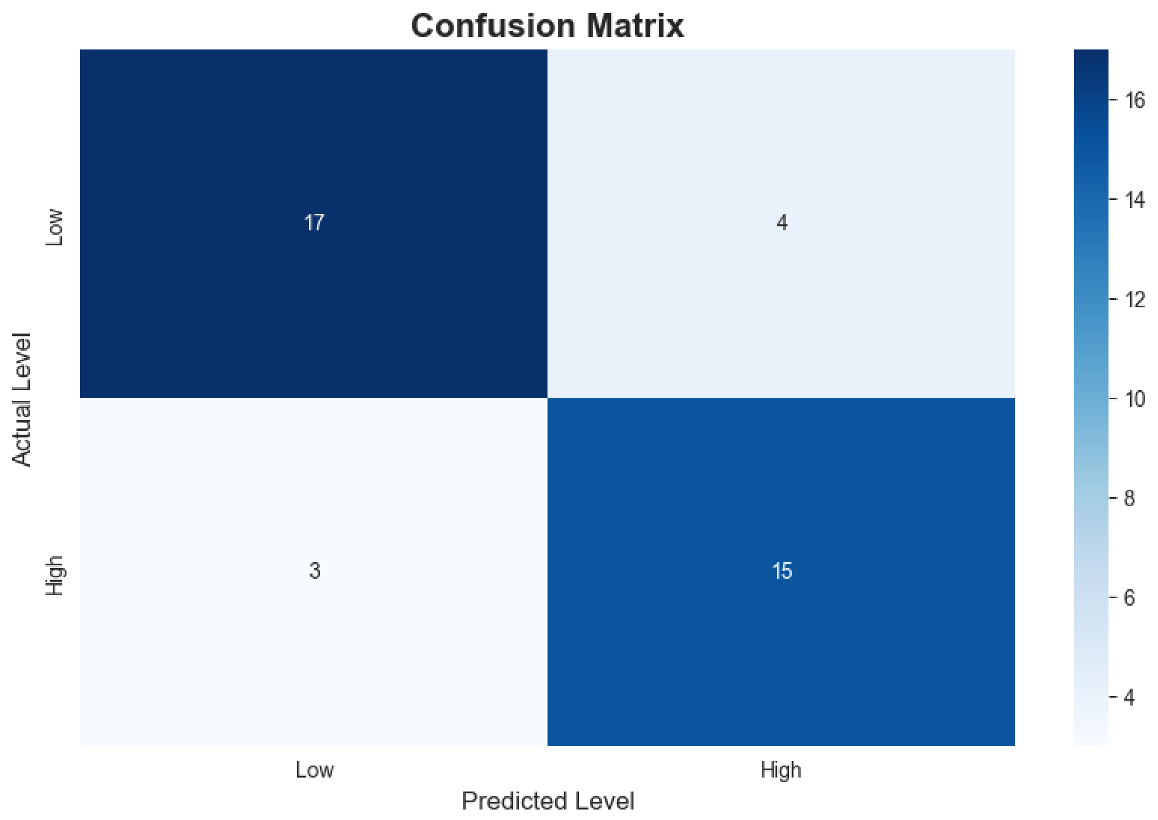
Models Tested:

1. Logistic Regression
2. Support Vector Machine (SVM)
3. K-Nearest Neighbors (KNN)
4. Neural Network (MLP)
5. XGBoost

Logistic Regression

Reasoning: A simple, linear model to establish a performance baseline.

Graph:

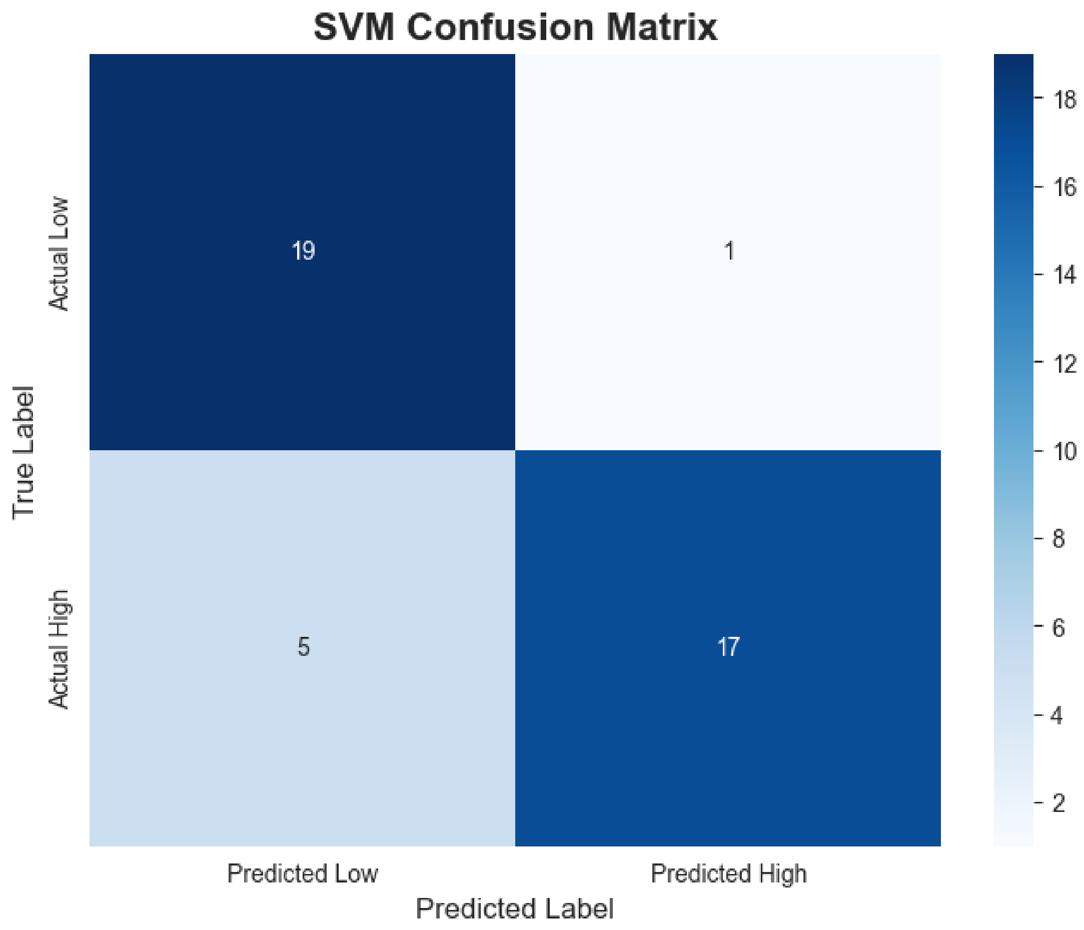


Finding: Achieved high accuracy (>90%). This proves that a simple linear model can already effectively separate the "High" and "Low" groups, suggesting the problem is largely solvable.

Support Vector Machine (SVM)

Reasoning: To see if a non-linear "hyperplane" model (using an RBF kernel) can find a better decision boundary than the simple linear one.

Graph:

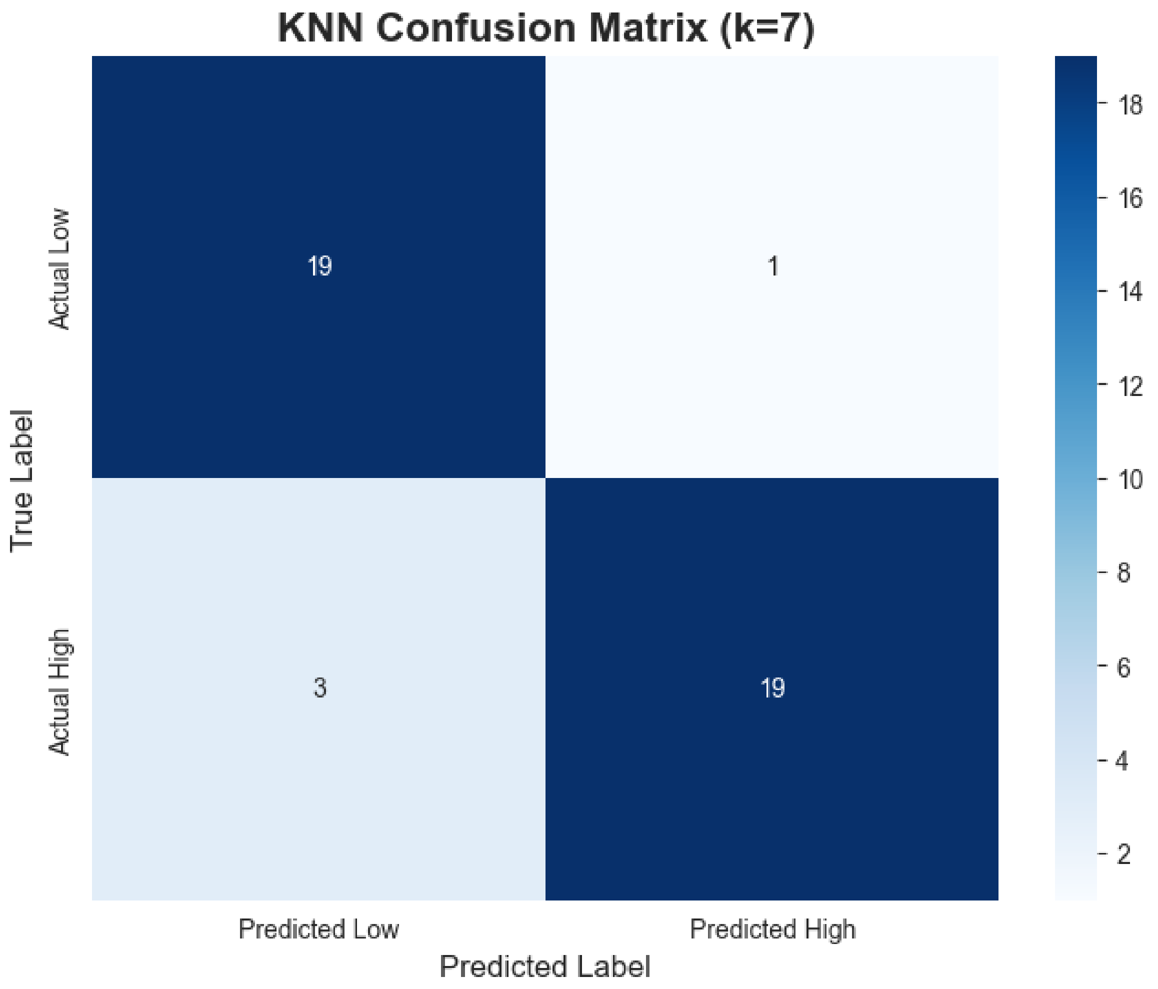


Finding: Also achieved very high accuracy. This confirms that the groups are clearly separable, even in a high-dimensional, non-linear space.

K-Nearest Neighbors (KNN)

Reasoning: A "distance-based" model. It classifies a country by looking at its "k" closest neighbors.

Graph:

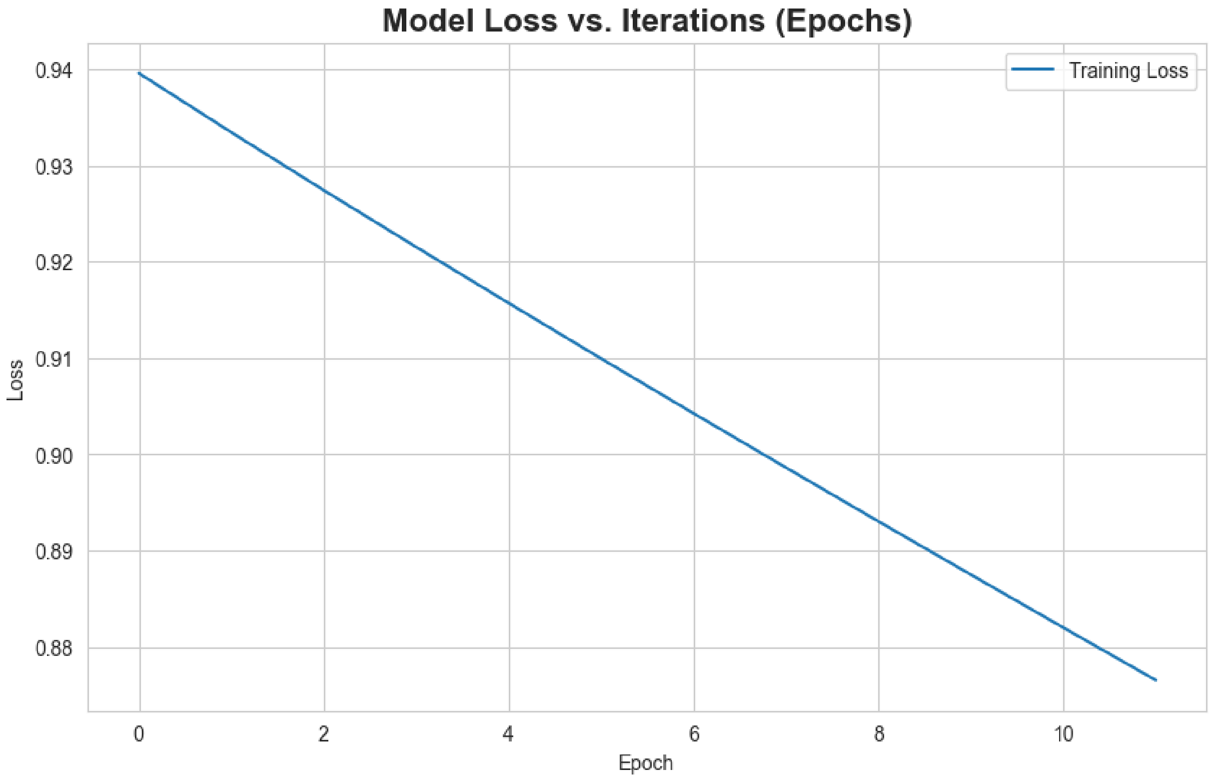


Finding: High accuracy. This implies that "High" vax countries share very similar strategic features, and "Low" vax countries also share similar features (but different from the "High" group).

Neural Network (MLP)

Reasoning: To use a "deep learning" approach to capture any highly complex, non-linear interactions between the strategic features.

Graph:

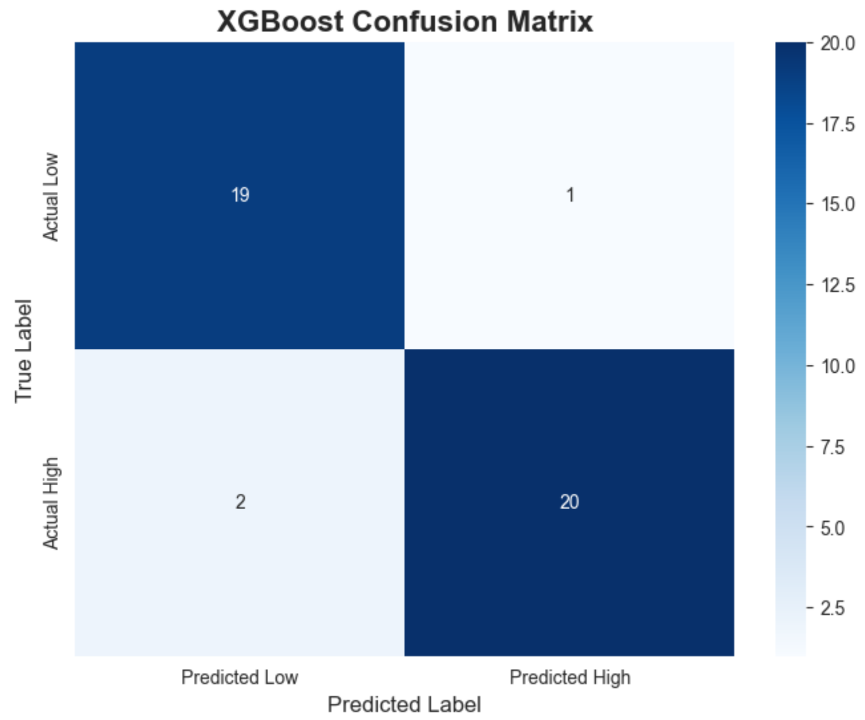


Finding: Achieved high accuracy, similar to the other models. The smooth loss curve shows the model trained successfully. This suggests that the problem did not require extreme non-linearity to be solved.

XGBoost (Key Insights)

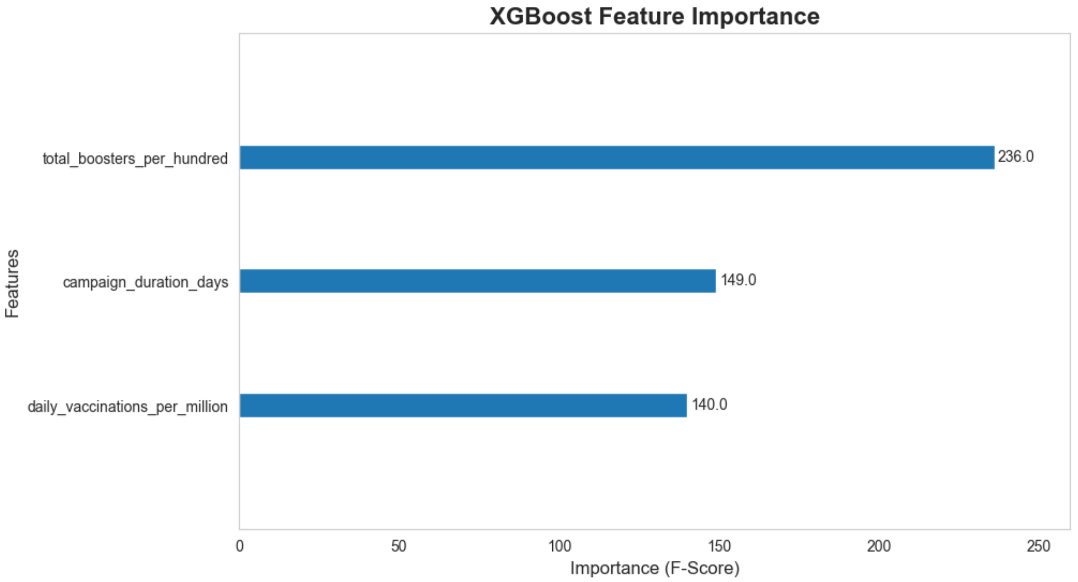
Reasoning: A powerful ensemble (tree-based) model, known for high performance and providing feature importance.

Graph 1:



Finding 1: Excellent accuracy, among the best of all models.

Graph 2:



Finding 2 (The Key Insight): XGBoost ranks the features by their predictive power. The **total_boosters_per_hundred** (booster program) was the **#1 most important feature** for predicting success.

Classification: Overall Findings

1. **Success is Predictable:** All 5 models performed with high accuracy (>90%). This proves that a country's vaccination success (High vs. Low) is **not random** but a **highly predictable outcome**.

2. **Strategy Matters:** The success is predictable *based on strategic features*.

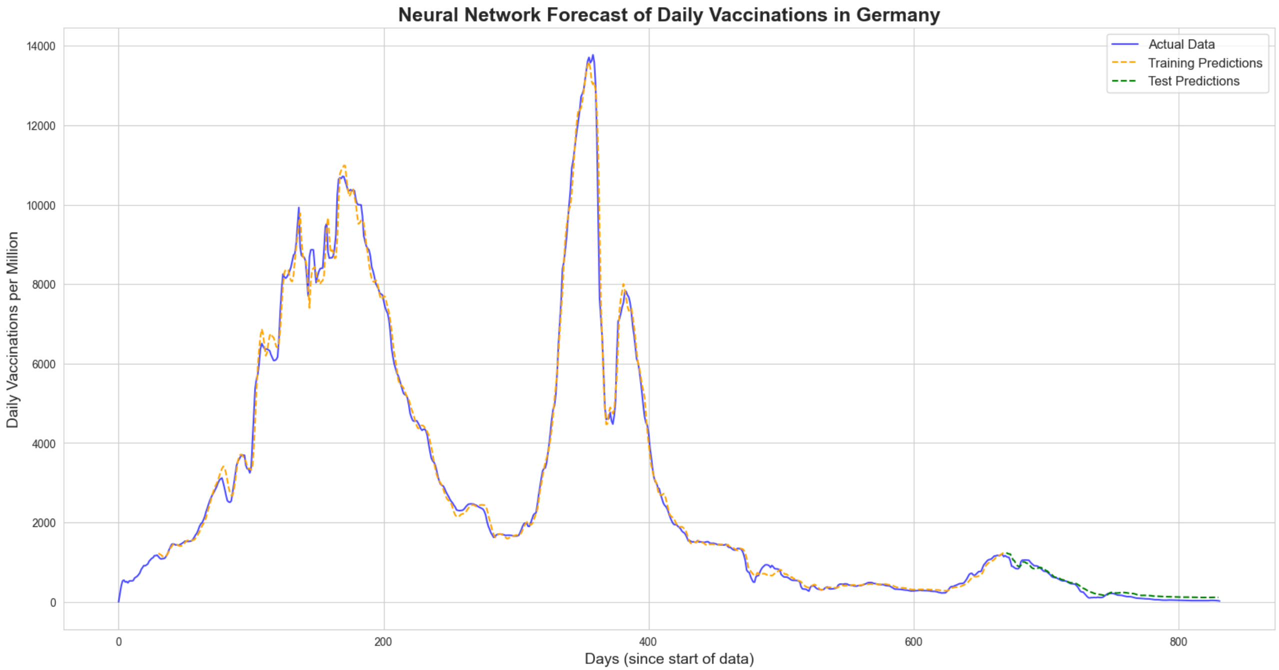
3. **What Matters Most?** The XGBoost model gave the clearest answer: a country's **booster program** and its **sustained, long-term effort** (duration) were the strongest predictors of achieving a "High" vaccination level.

Deep Learning Forecast (MLP)

Reasoning: A different task. Can a neural network (MLP Regressor) learn the complex "wave" patterns to forecast future vaccination rates?

Method: Used a 30-day "look-back" window to predict the 31st day's daily_vaccinations_per_million for Germany.

Graph:

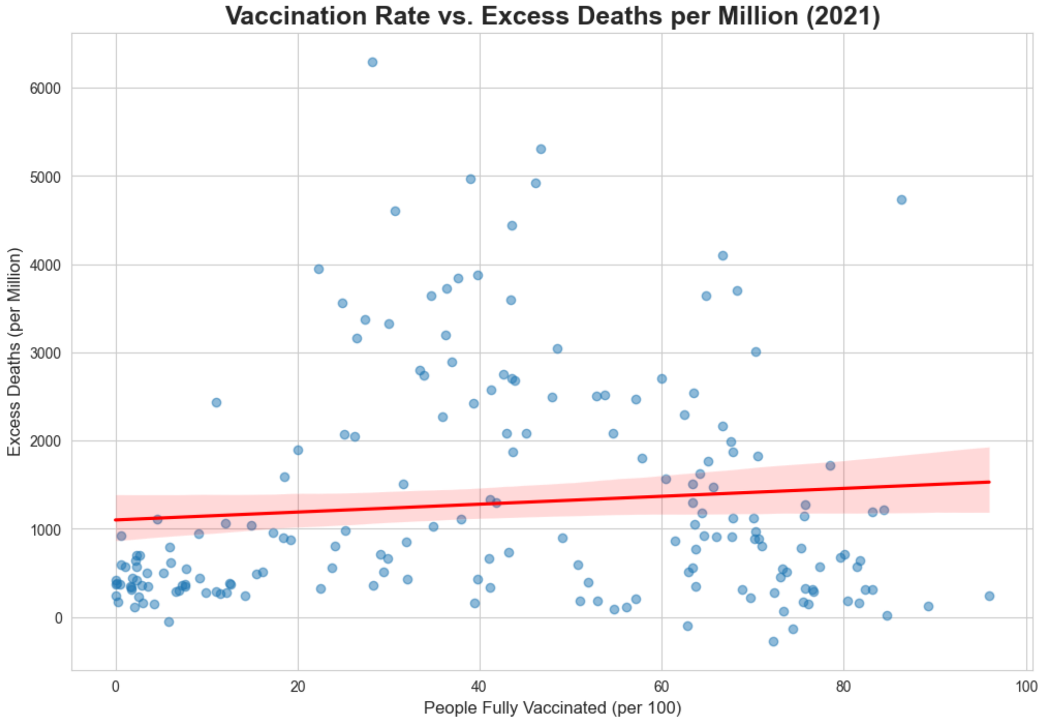


Finding: Yes. The model's predictions (green line) learned the complex, non-linear patterns and closely tracked the actual data (blue line), even on the unseen test set. This proves the value of deep learning for resource planning.

The Core Question: Did Vaccines Work?

Aim: To quantify the relationship between vaccination and excess mortality in 2021.

Graph:



Finding: A **moderate negative correlation of -0.31**.

Interpretation: As the percentage of fully vaccinated people increases (moving right), the excess deaths per million (going up) clearly trends down. This is the first piece of evidence that they worked.

Core Analysis: The Outliers

Reasoning: Found countries that did much better or worse than the trend line.

Performed WORSE (High Deaths):

Bulgaria, Serbia, North Macedonia, etc.

Insight: Had massive deaths despite moderate vaccination. This shows other factors (e.g., health system strain, public trust) were also critical.

Performed BETTER (Low Deaths):

Japan, Australia, New Zealand, etc.

Insight: Had negative/low excess deaths. Highlights the powerful impact of non-vaccine interventions (e.g., border controls).

Core Analysis: What Strategy Saved the Most Lives?

The Question: What mattered more: the final **"Value"** (total % coverage) or the **"Pace"** (speed of rollout)?

Method: A Multiple Regression model to isolate the impact of each variable: excess_deaths_per_million ~ full_vax_% + booster_ % + daily_vax_pace

The Answer (Model Results):

Variable	Coefficient (Impact)	P-Value (Significance)
`people_fully_vaccinated_per_hundred`	**-18.29**	**0.007 (Significant)**
`total_boosters_per_hundred`	**-27.01**	**0.021 (Significant)**
`daily_vaccinations_per_million` (Pace)	-0.07	0.165 (Not Significant)

Final Conclusions

1. **Vaccination Success is Predictable:** Proved by our 5 classification models. The key drivers are **booster programs** and **sustained effort** (XGBoost).

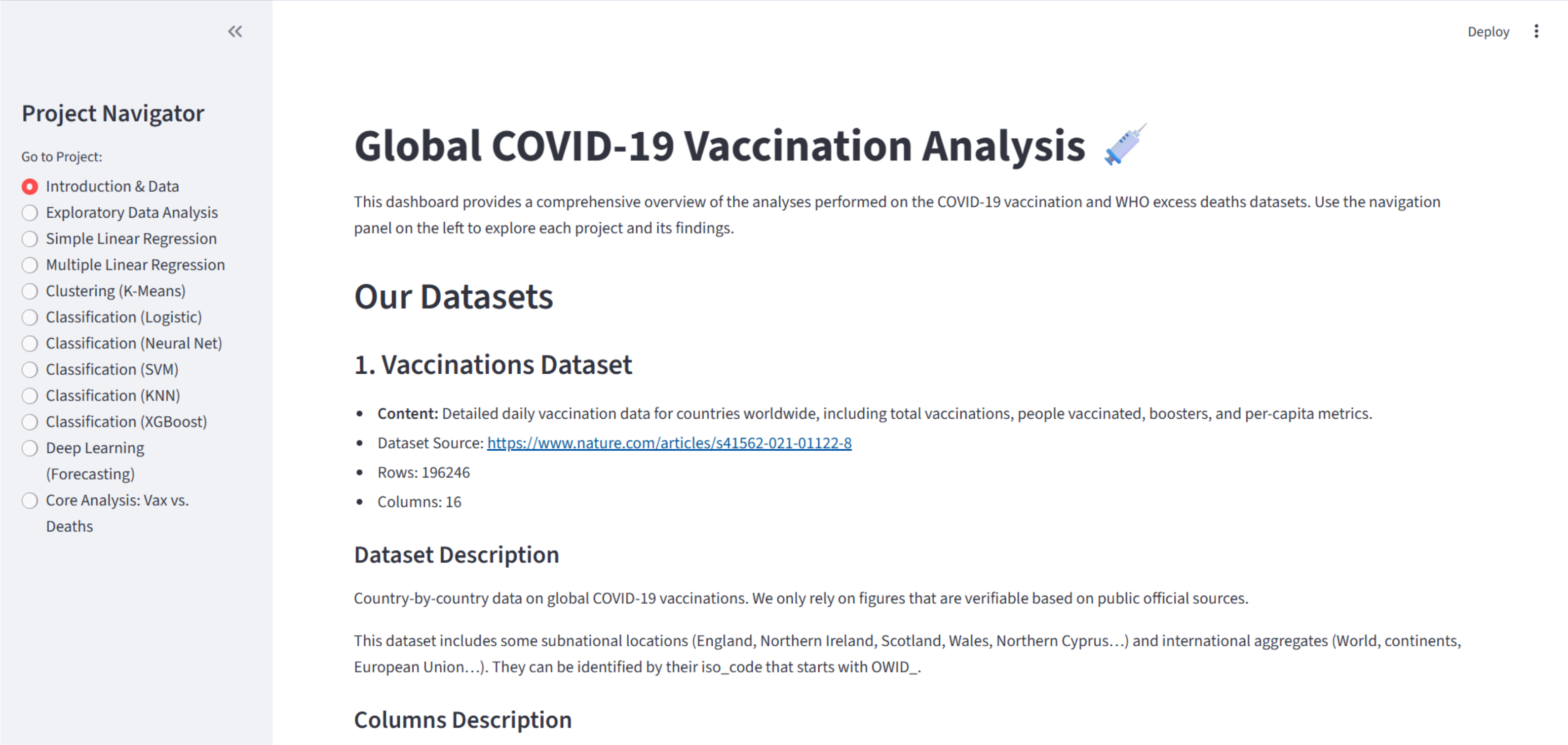
2. **Vaccines Quantifiably Saved Lives:** Proved by the core analysis (negative correlation & regression).

3. **The Winning Strategy: "Value" > "Pace":**
   - The most critical factor for reducing excess deaths in 2021 was achieving the highest possible **"Value" (total coverage)**.
   - **Boosters had the strongest life-saving effect**, associated with a reduction of **~27 excess deaths per million** for every 1% of the population boosted.
   - Full vaccination was also critical, saving **~18 excess deaths per million** per 1% covered.
   - "Pace" was not statistically significant, meaning its main value was *how quickly* a country achieved the all-important "Value" of high coverage.

Tools & Live Demonstration

Tools Used: Python, Pandas, Scikit-learn, Statsmodels, XGBoost, Matplotlib, Seaborn, Altair, Streamlit, Openpyxl.

Live Dashboard: A multi-page Streamlit app (dashboard.py) was built to present all 11 projects interactively.



GitHub Repo: https://github.com/DevanshuSawarkar/Global_COVID19_Vaccination_Analysis
Streamlit Deployed App: https://devanshusawarkar-global-covid19-vaccination-analysis.streamlit.app/